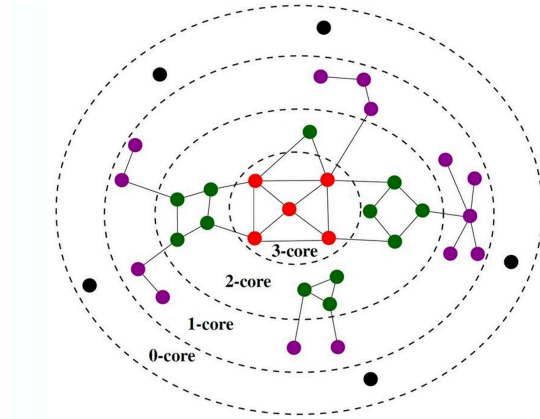


L6: k-core Decomposition

In some applications of network analysis, instead of trying to *rank* individual nodes in terms of centrality, we are interested in identifying the most important group of nodes in the network. There are different ways to think about the importance of groups of nodes. One of them is based on the notion of **k-core**:

A *k-core* (or “core of order- k ”) is a maximal subset of nodes such that each node in that subset is connected to at least k others in that subset.

Note that, based on the previous definition, a node in the k -core also belongs to all lower cores, i.e., to any k' -core with $k' < k$.



(Image from research paper: [Perturb and combine to identify influential spreaders in real-world networks](https://dl.acm.org/doi/abs/10.1145/3341161.3342866), [AJP Tixier, MEG Rossi, FD Malliaros, J Read](https://dl.acm.org/doi/abs/10.1145/3341161.3342866))

The visualization shows a network in which the red nodes belong to a core of order-3 (the highest order in this example), the green and red nodes form a core of order-2, while all nodes (except the black) form a core of order-1.

Note that there is a purple node with degree=5 – but its highest order is 1 because all its connections except one are to nodes of degree-1. Similarly,

there are green nodes of degree-3 that are in the 2-core set.

A simple algorithm, known as “**k-core decomposition**”, can associate each node with the highest k -core order that that node belongs to. The algorithm proceeds iteratively by removing all nodes of degree k , for $k=0, 1, 2, \dots$. During the iteration that the algorithm removes nodes of degree k , some higher-degree nodes may lose connections and become of degree- k . Those nodes would also be removed, joining the k -core until no further nodes can be removed in that iteration. The algorithm terminates when all nodes have been removed. Note that the maximum-order core set may not form a connected component.

One way to think about k -cores is as the successive layers of an onion, where $k=1$ corresponds to the external layer of the onion, while the highest value of k corresponds to the “heart” of the onion. Using this metaphor, the k -core decomposition process gradually peels off the network layer by layer until it reaches its most internal group of nodes.

The nodes in the maximum core order are considered as the most well-connected group of nodes in the network.

Additionally, the k -core decomposition process is useful by itself because it allows us to assign a layer to each node, with nodes of lower order (lower values of k) considered as peripheral, and nodes of higher-order as more central in the overall network.

A related concept is:

- The **k-shell** is the subgraph induced by nodes in the k -core that are not in the $(k+1)$ -core. For example, the green nodes in the visualization form the 2-shell.

Food For Thought

We claim that the k -core decomposition approach can rank nodes in terms of importance, and that the highest-order core consists of the most important nodes. Can you come up with an example in which this claim is probably true -- and maybe another example in which it is not?